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SCHOOL FUTURE TECH PROGRAMM

Faculty Name :

Course Duration	: 5 Months (150 Days)
Classes	: Offline & Online
Daily Teaching	: 3 Hours
Total Hours	: 300 Hours

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LEVEL 1 – JUNIOR INNOVATORS (5th–6th Standard)

MODULE 1 – COMPUTER BASICS (Classes 1–20)

Class 6 — Storage Devices (Complete Notes)

Complete Theory & Practical Notes

Learning Objectives:-

By the end of this class, students will be able to:-

- Understand what storage devices are.
- Identify different types of storage devices.
- Explain the uses of Hard Disk, SSD, USB Flash Drive, and Memory Card.
- Compare different storage devices.
- Perform a practical activity using storage devices.
- Create an interactive Scratch project about storage devices.

What are Storage Devices?







A **Storage Device** is a device used to **save, store, and retrieve data** such as documents, photos, videos, music, games, and software.

Without storage devices, a computer cannot permanently save your work.

Examples of Data Stored

- Documents
- Photos
- Videos
- Music
- Games
- School Projects

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Types of Storage Devices

There are many storage devices, but today we will learn about four common ones:

1. Hard Disk Drive (HDD)
2. Solid State Drive (SSD)
3. USB Flash Drive (Pen Drive)
4. Memory Card (MicroSD Card)

1. Hard Disk Drive (HDD)







What is a Hard Disk?

A ***Hard Disk Drive (HDD)*** stores large amounts of data inside a computer using magnetic disks.

Features

- Large storage capacity
- Lower cost
- Stores Operating System and files
- Mechanical moving parts

Uses

- Store school projects
- Save videos and photos

- Install software
- Store games

Advantages

Large storage

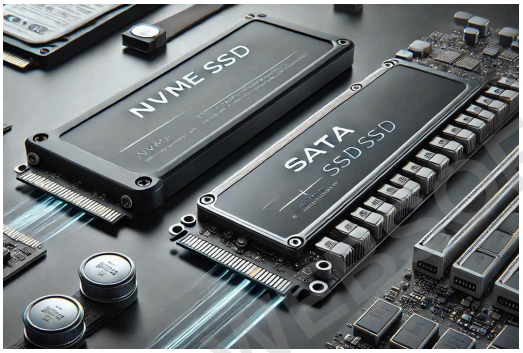
Affordable

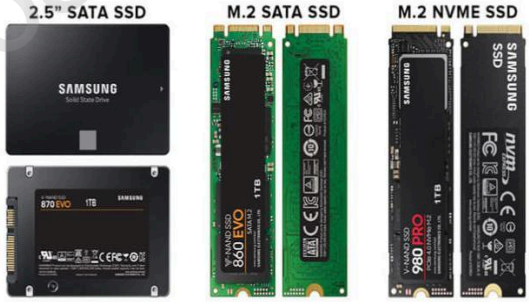
Disadvantages

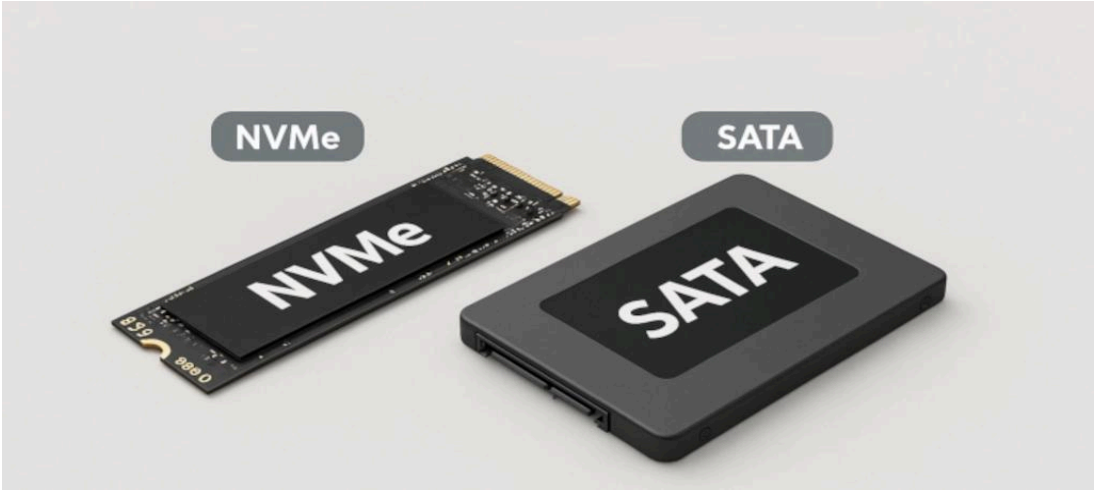
Slower than SSD

Can be damaged by strong shocks

2. Solid State Drive (SSD)







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What is an SSD?

A ***Solid State Drive (SSD)*** stores data using electronic memory chips. It has **no moving parts**, making it faster and more durable than an HDD.

Features

- Very fast
- Lightweight
- Silent operation
- Energy efficient

Uses

- Fast computer startup
- Gaming
- Programming
- Video editing

Advantages

Fast performance

Durable

Disadvantages

More expensive than HDD

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3. USB Flash Drive (Pen Drive)







What is a USB Flash Drive?

A ***USB Flash Drive***, also called a ***Pen Drive***, is a portable storage device used to carry files from one computer to another.

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Features

- Small and lightweight
- Easy to carry
- Plug and Play
- Reusable

Uses

- Transfer files
- Backup school work
- Store presentations
- Carry photos and videos

Advantages

Portable

Easy to use

Disadvantages

Easy to lose due to small size

4. Memory Card



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What is a Memory Card?

A ***Memory Card*** is a small storage device commonly used in mobile phones, tablets, cameras, and drones.

Features

- Very small
- Lightweight
- Different storage sizes
- Portable

Uses

- Store photos
- Save videos
- Store music
- Expand phone storage

Advantages

Fits inside small devices

Disadvantages

Easy to misplace

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Comparison of Storage Devices

Device	Portable	Speed	Common Use
Hard Disk (HDD)	No	Medium	Desktop Computers
SSD	Usually Internal	Very Fast	Laptops & PCs
USB Flash Drive	Yes	Fast	File Transfer
Memory Card	Yes	Fast	Phones & Cameras

Storage Device Speed Comparison

Relative Speed of Storage Devices

Illustrative comparison of common storage devices.

036912Hard DiskSSDUSBMemory Card

Easy Memory Trick

Device	Remember It Like
HDD	Big Storage Box
SSD	Super Fast Rocket
USB	Pocket File Carrier
Memory Card	Tiny Memory Chip

Classroom Practical Activity

Activity 1 – Identify the Storage Device

The teacher shows real storage devices or pictures.

Students answer:-

- What is its name?
- Is it portable?
- Where is it used?
- What can it store?

Activity 2 – Match the Device

Device	Used In
HDD	Desktop Computer
SSD	Laptop
USB	File Transfer
Memory Card	Mobile Phone & Camera

Activity 3 – Storage Hunt

Ask students to find storage devices they have seen at home, such as:

- Laptop SSD

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- Desktop Hard Disk
- Pen Drive
- Mobile Memory Card

Discuss where each is used.

Practical Coding Activity (Scratch)

Project Title

"Know Your Storage Devices"

Objective

Students create an interactive Scratch project where clicking each storage device displays its name and purpose.

Step 1

Open ***Scratch*** and create a new project.

Delete the default Cat sprite.

Step 2

Add Sprites

- Hard Disk
- SSD
- USB Flash Drive
- Memory Card

Choose a computer lab background.

Step 3 – Programming

Green Flag

when green flag clicked
say "Welcome! Click on a storage device to learn about it." for 3 seconds

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Hard Disk Sprite

when this sprite clicked
say "I am a Hard Disk. I store large amounts of data." for 4 seconds

SSD Sprite

when this sprite clicked
say "I am an SSD. I am faster than a Hard Disk." for 4 seconds

USB Flash Drive Sprite

when this sprite clicked
say "I am a USB Flash Drive. I help transfer files." for 4 seconds

Memory Card Sprite

when this sprite clicked
say "I am a Memory Card. I store photos and videos in phones and cameras." for 4 seconds

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Advanced Scratch Challenge

Create a quiz with a ***Score*** variable.

Example:-

ask "Which storage device is the fastest?" and wait

If answer =

SSD

Then

change Score by 1
say "Excellent!" for 2 seconds

Else

say "Try Again!" for 2 seconds

Add More Questions

- Which device is commonly used inside desktop computers?
- Which device is used to transfer files between computers?
- Which storage device is commonly used in mobile phones?

- Which device has no moving parts?

Display the total score at the end.

AI Connection

Discuss with students:-

- Where are AI apps and games stored on a computer?
- Why do AI computers often use fast SSDs?
- Why do phones use memory cards to store photos and videos?
- How do storage devices help save our school projects and AI creations?

Fun Quiz:-

1. Which storage device is the fastest?

- A) Hard Disk
- **B) SSD**
- C) Memory Card

2. Which device is also called a Pen Drive?

- A) Hard Disk
- **B) USB Flash Drive**
- C) SSD

3. Which storage device is commonly used in mobile phones?

- A) Hard Disk
- B) SSD
- **C) Memory Card**

4. Which storage device has moving parts?

- **A) Hard Disk**
- B) SSD
- C) USB Flash Drive

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5. Which device is best for carrying files from one computer to another?

- A) Memory Card
- **B) USB Flash Drive**
- C) Hard Disk

Homework:-

1. Draw and label the four storage devices.
2. Write ***two uses*** of each storage device.
3. Create a comparison table showing ***Portability, Speed,*** and ***Common Use***.
4. Improve your Scratch project by adding:
    - Sounds
    - Animations
    - A score variable
    - A "Congratulations!" message after completing the quiz.
5. At home, identify which storage devices are used in your family's computer or mobile phone.

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Learning Outcomes:-

After completing this lesson, students will be able to:-

- Define what a storage device is.
- Identify the four common storage devices: ***Hard Disk, SSD, USB Flash Drive, and Memory Card.***
- Explain the features, advantages, and uses of each storage device.
- Compare storage devices based on speed and portability.
- Build an interactive Scratch project to teach others about storage devices.
- Understand the importance of storage devices in saving files, software, photos, videos, and school projects.